NUI Galway
OÉ Gaillimh

Semester I Examinations 2019/2020

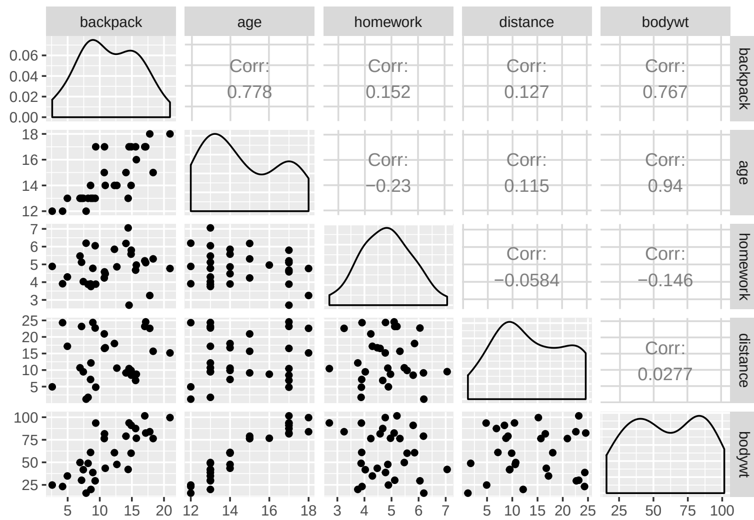
Exam Codes	2BDS1, 3BA1, 3BS9, 3MR3, 3BPT1,4BA4, 4BME1, 4BS2, 1CNS1, 1CSD1, 1OA1, 1OA2, 1EM1
Exams	Second Arts, Third Arts, Third Science, Third Marine Science, Fourth Arts, Fourth Mathematics and Education, Fourth Science, Master of Science
Module	Applied Statistics I
Module Codes	**ST311**
External Examiner	Professor S. Wilson
Internal Examiner(s)	Dr. E. Holian Professor J. Newell
Instructions:	**Answer ALL SIX Questions**
Duration	2 Hours
No. of Pages	13 pages, including this one
School	School of Mathematics, Statistics and Applied Mathematics
Release to Library:	Yes ☑ No ☐
Release in Exam Venue	Yes ☑ No ☐

Requirements:

Statistical Tables/ Log Tables	Relevant distribution tables are attached to this paper. The *New Formulae & Tables* are optional.
Other Materials Allowed	A calculator is allowed (non-programmable and not capable of storing text).

Researchers running the School Years Welfare Study were interested in explaining the variability in weights of backpacks, `backpack`, carried by children going to school. The following candidate predictor variables were recorded for a sample of 30 children:

- `age` : the age of the child in years
- `bodywt` : bodyweight in kgs.
- `homework` : number of hours homework per week
- `distance` : distance between home and school in kilometers.



Answer the following questions:

Question 1

(a) *"The true pearson correlation coefficient between two variables X and Y for a population of individuals is known to have value $\rho = 0$. This implies that X and Y are unrelated."*
Is this statement true or false? If you deem the statement to be false give a reason for your answer. [2 marks]

(b) The sample of 30 individuals gave a sample pearson correlation coefficient of $r = 0.778$ for backpack and age. Interpret this value. [2 marks]

(c) Carry out pearson's correlation test to determine whether there is evidence of a significant increasing relationship between the variables backpack and age in the population. Include clear statements of the hypotheses, the rejection region and conclusion. [6 marks]

[Total 10 marks]

Question 2

The following model is being considered in order to model the variability in the response variable *backpack*, using three candidate predictor variables, *age*, *homework* and *distance*,

$$backpack_i = \beta_0 + \beta_1 age_i + \beta_2 homework_i + \beta_3 distance_i + \epsilon_i.$$

(a) What does the error term, ϵ_i, represent? [2 marks]

(b) When carrying out an Analysis of Variance, explain in general what the terms SS_{Total}, SS_{Reg} and SS_{Res} represent. [5 marks]

(c) The model is fitted to observations obtained for a sample of 30 individuals, and the results of the fit are provided in the partial printout below.

```
Coefficients:
               Estimate  Std. Error   t value   Pr(>|t|)
(Intercept)    -26.2531    4.8922    -5.3663    0.0000
age              2.0009    0.2503    7.9929    0.0000
homework         1.6775    0.5071      -          -
distance         0.0318    0.0666    0.4775    0.6370
---
Residual standard error:  -  on  -  degrees of freedom
Multiple R-squared:   -   ,    Adjusted R-squared:  -
F-statistic: 22.6  on  -  and  -  DF,  p-value: 0.0000002054334
```

Analysis of variance for the model fitted to the observed data gives the values $SS_{Total} = 611.18$ and $SS_{Reg} = 441.78$.

(i) The coefficient of determination R^2 is missing from the output, calculate this value and interpret the meaning of this value in this application. [2 marks]

(ii) The value of the residual standard error, s, is missing from the output. Find this value. [3 marks]

(iii) The test statistic for the ANOVA F-test is printed as 22.6. Show how this value is calculated. Write down the hypotheses being tested. Referencing the p-value provided, what is the conlcusion for this test in terms of the variables in the application. [6 marks]

(iv) Complete the missing t-test in the output, i.e. test whether homework is a good predictor of backpack at the $\alpha = 0.05$ level. Include the hypotheses being tested, the calculation of the test statistic, a rejection region, a decision and conclusion. [5 marks]

(d) The output below provides a prediction interval and a confidence interval for the response when $age = 14$, $homework = 5$ and $distance = 11$.
Prediction interval for the response:

```
       fit      lwr      upr
1 10.49695 5.142898 15.85099
```

Confidence interval for the response:

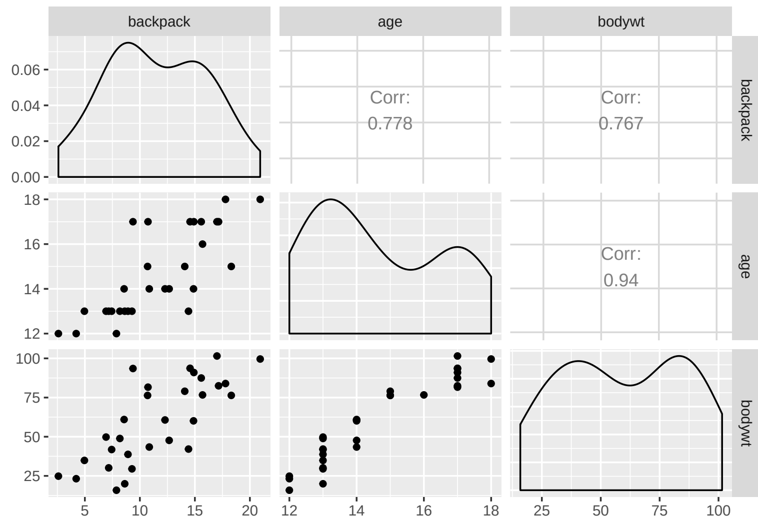
```
       fit      lwr      upr
1 10.49695 9.430513 11.56338
```

Explain the difference as to when a prediction interval and a confidence interval applies. [2 marks]

[Total 25 marks]

Question 3

The plot and output below is produced to assess a model where age and bodywt are used to predict backpack weight, $backpack_i = \beta_0 + \beta_1 age_i + \beta_2 bodywt_i + \epsilon_i$.




```
Call:
lm(formula = backpack ~ age + bodywt, data = Data)

Residuals:
    Min      1Q  Median      3Q     Max
-6.6914 -1.9923  0.4405  1.6110  5.6846

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -8.42285    8.70075  -0.968    0.342
age          1.14748    0.82034   1.399    0.173
bodywt       0.05324    0.06059   0.879    0.387

Residual standard error: 2.949 on 27 degrees of freedom
Multiple R-squared:  0.6157,    Adjusted R-squared:  0.5873
F-statistic: 21.63 on 2 and 27 DF,  p-value: 0.00000247
```

Discuss the predictive power of the two variables, *age* and *bodywt*, by giving your conclusions from the outputs of the t-tests and ANOVA test provided. Ensure you provide the hypotheses being tested for any test that you reference.

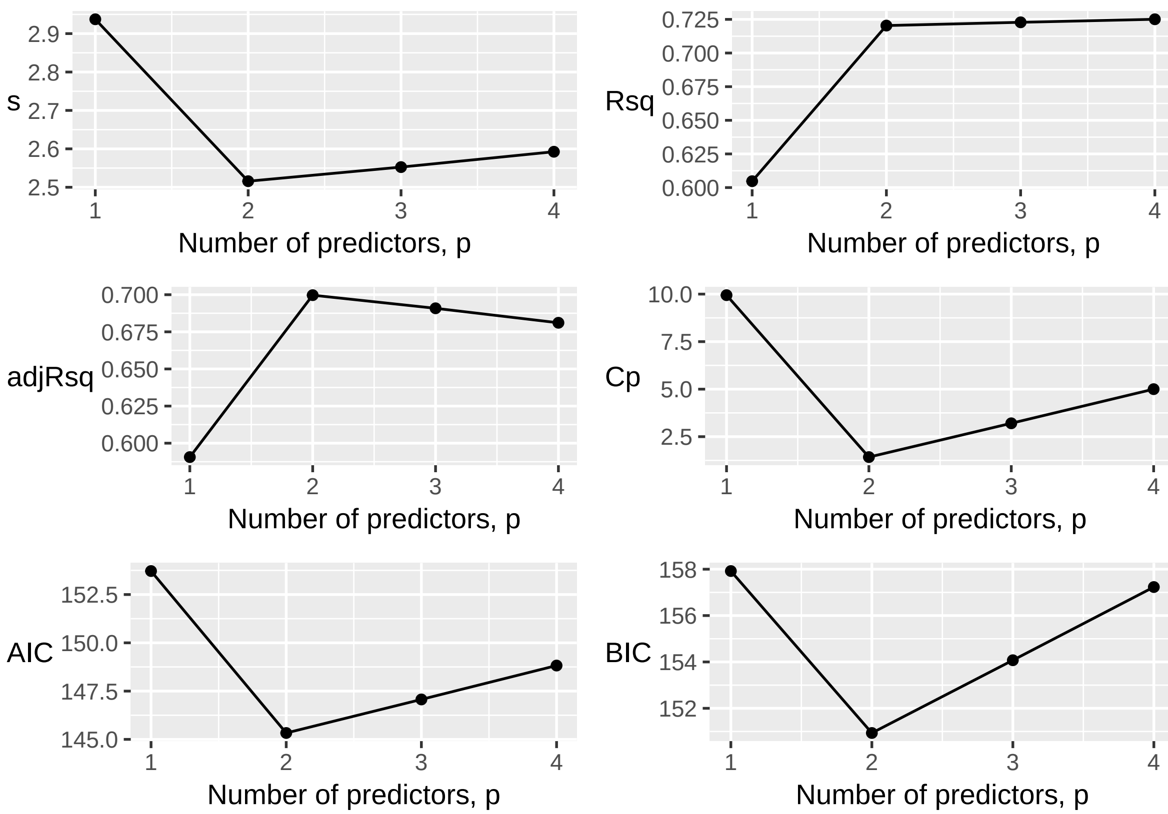
[Total 15 marks]

Question 4

(a) Explain the meaning of the phrase "a parsimonious model". [2 marks]

(b) A best-subsets routine was carried out on 4 candidate predictors, age, homework, distance and bodywt. The best fitting model for each number of predictors is selected and summarised below.

model	predictors	p	s	Rsq	adjRsq	Cp	AIC	BIC
1	age	1	2.937	0.605	0.591	9.944	153.716	157.920
2	age homework	2	2.516	0.720	0.700	1.426	145.330	150.935
3	age homework distance	3	2.553	0.723	0.691	3.204	147.068	154.074
4	age homework distance bodywt	4	2.592	0.725	0.681	5.000	148.824	157.231



Using the output to support your answer, write down the most suitable model, i.e. the set of predictors, that you think best explains the variability in observed `backpack`. Justify your choice of model by explaining the role of s, R^2, adjusted R^2, and Mallow's C_p in choosing the best model. [8 marks]

[Total 10 marks]

Question 5

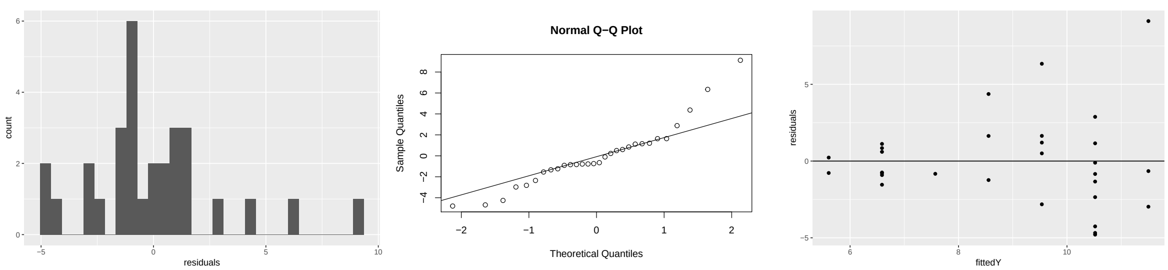
The percentage load of weight of the backpack, `PercLoad`, is calculated as the backpack weight relative to the individual's body mass. The following output outlines the fit for the predictor age on the response `PercLoad`, i.e. $PercLoad = \beta_0 + \beta_1 age_i + \epsilon_i$.

```
Call:
lm(formula = PercLoad ~ age, data = Data)

Residuals:
    Min      1Q  Median      3Q     Max
-4.7928 -1.3114 -0.6993  1.1469  9.1218

Coefficients:
            Estimate Std. Error t value   Pr(>|t|)
(Intercept)  23.2933     4.2454   5.487 0.00000735 ***
age          -0.9826     0.2876  -3.416    0.00196 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.03 on 28 degrees of freedom
Multiple R-squared:  0.2942,    Adjusted R-squared:  0.269
F-statistic: 11.67 on 1 and 28 DF,  p-value: 0.001959
```



(a) What assumptions are required for regression analysis? [4 marks]

(b) An observation with predictor value $age = 12$ had observed response $PercLoad = 20.62$. Find the value of the fitted response $\widehat{PercLoad}$ and the fitted residual $\hat{\epsilon}$ for this individual, and thus show how the standardised residual of 3.01 is arrived at.

Why would this observation be considered an unusual observation? [4 marks]

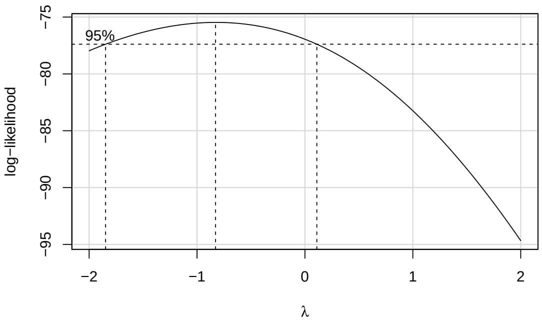
(c) Explain how the diagnostic plots are used to assess validity of the assumptions. [4 marks]

(d) What does the following box-cox transformation output indicate?

```
bcPower Transformation to Normality
   Est Power Rounded Pwr Wald Lwr bnd Wald Upr Bnd
Y1   -0.8236           0       -1.798       0.1507

Likelihood ratio tests about transformation parameters
                              LRT df         pval
LR test, lambda = (0)  2.959074  1 0.0853968227
LR test, lambda = (1) 15.545215  1 0.0000805553
```

Question 5 continues on the next page



[4 marks]

(e) The following output fits the same model but with a log transformation on the response variable, i.e. $log(PercLoad) = \beta_0 + \beta_1 age_i + \epsilon_i$.

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   3.7062     0.4123  8.9886  <0.0001
age          -0.1083     0.0279 -3.8752   0.0006
```

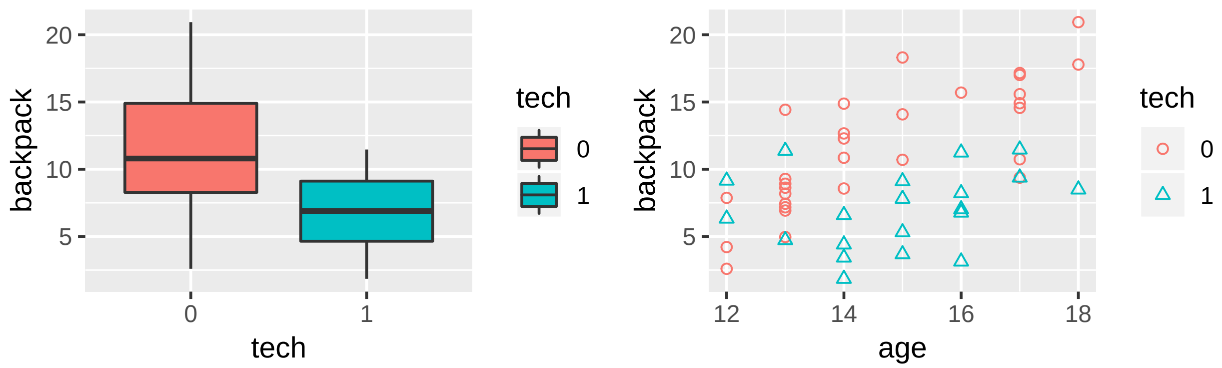
Use this model to predict PercentLoad for a student aged 12 years. [4 marks]

[Total 20 marks]

Question 6

In addition to the previous data which referred to children who use no technology in their school day, these variables were also recorded children who said a tablet was used as part of their school day. The variable *tech* is stored as an indicator variable defined as

$$tech_i = \begin{cases} 0 & \text{if child } i \text{ uses no technology} \\ 1 & \text{if child } i \text{ uses technology, e.g. tablet} \end{cases}$$



(a) The model $backpack_i = \beta_0 + \beta_1 tech_i + \epsilon_i$, is fitted to the data, giving:

```
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  16.1374     1.7126  9.4225  <0.0001
tech         -4.5821     1.1547 -3.9683   0.0002
```

(i) Use the fitted model to predict backpack weight for:
- a child that doesn't use technology.
- a child that does use technology. [2 marks]

(ii) Interpret the meaning of the estimated coefficient of *tech* and comment on the significance of the effect. Ensure you state the hypotheses being tested when referring to the results of the output. [3 marks]

(iii) This analysis is equivalent to which of the following two population comparison tests:
- Paired t-test for difference in population means
- Independent sample t-test for difference in population means with equal variance
- Independent sample t-test for difference in population means with unequal variance
- Wilcoxon rank sum test for difference in population medians [1 mark]

(b) A plot of the observed relationship of backpack weights with age is provided, with an indication to those in the sample that did or didn't use technology in their school day. The age of each child is added to the model as a covariate,
$backpack_i = \beta_0 + \beta_1 age_i + \beta_2 tech_i + \epsilon_i$, giving the output:

```
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  -3.7314     3.7552 -0.9937   0.3255
age           1.3830     0.2443  5.6606  <0.0001
tech         -4.9509     0.9022 -5.4878  <0.0001
```

Question 6 continues on the next page

(i) Interpret the meaning of the fitted coefficients of *age* and *tech* in this model. [3 marks]

(ii) For this model, write down the estimated relationship between backpack and age for:

- a child that doesn't use technology.
- a child that does use technology. [2 marks]

(iii) What assumption is being made about the relationship of backpack weight and age in this model? [2 marks]

(c) An interaction term, $age : tech$, calculated as $(age_i) \times (tech_i)$ for each individual, is added to the model,

$$backpack_i = \beta_0 + \beta_1 age_i + \beta_2 tech_i + \beta_3 age : tech_i + \epsilon_i$$

giving the output:

```
            Estimate Std. Error  t value Pr(>|t|)
(Intercept) -30.7785    10.1458  -3.0336   0.0040
age           3.2140     0.6837   4.7008  <0.0001
tech         15.6276     7.2938   2.1426   0.0375
age:tech     -1.3890     0.4890  -2.8403   0.0067
```

(i) For this model, write down the estimated relationship between backpack and age for:

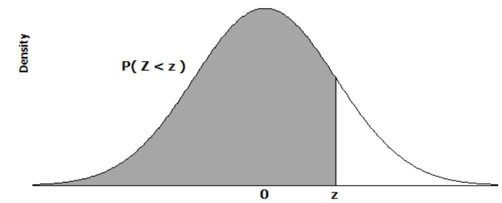
- a child that doesn't use technology.
- a child that does use technology. [2 marks]

(ii) Interpret the meaning of the fitted coefficient of $age : tech$ in this model. Determine if the term is significant, using the result of the individual t-test in this analysis. Ensure you state the hypotheses being tested when referring to the results of the output. [5 marks]

[Total 20 marks]

Table: Cumulative Probabilities for the Standard Normal Distribution, Z~Normal (0,1).

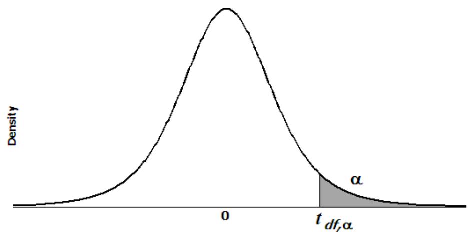
Selection of values from R command : pnorm(z, mean = 0, sd = 1, lower.tail = TRUE)



Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

Table: t distribution critical values

Selection of values from R command: qt(p, df, lower.tail = FALSE)

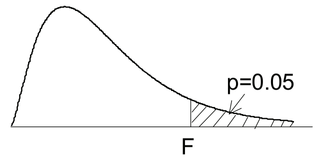


	Upper tail probability											
df	**0.25**	**0.20**	**0.15**	**0.10**	**0.05**	**0.025**	**0.02**	**0.01**	**0.005**	**0.0025**	**0.001**	**0.0005**
1	1.000	1.376	1.963	3.078	6.314	12.706	15.895	31.821	63.657	127.321	318.309	636.619
2	0.816	1.061	1.386	1.886	2.920	4.303	4.849	6.965	9.925	14.089	22.327	31.599
3	0.765	0.978	1.250	1.638	2.353	3.182	3.482	4.541	5.841	7.453	10.215	12.924
4	0.741	0.941	1.190	1.533	2.132	2.776	2.999	3.747	4.604	5.598	7.173	8.610
5	0.727	0.920	1.156	1.476	2.015	2.571	2.757	3.365	4.032	4.773	5.893	6.869
6	0.718	0.906	1.134	1.440	1.943	2.447	2.612	3.143	3.707	4.317	5.208	5.959
7	0.711	0.896	1.119	1.415	1.895	2.365	2.517	2.998	3.499	4.029	4.785	5.408
8	0.706	0.889	1.108	1.397	1.860	2.306	2.449	2.896	3.355	3.833	4.501	5.041
9	0.703	0.883	1.100	1.383	1.833	2.262	2.398	2.821	3.250	3.690	4.297	4.781
10	0.700	0.879	1.093	1.372	1.812	2.228	2.359	2.764	3.169	3.581	4.144	4.587
11	0.697	0.876	1.088	1.363	1.796	2.201	2.328	2.718	3.106	3.497	4.025	4.437
12	0.695	0.873	1.083	1.356	1.782	2.179	2.303	2.681	3.055	3.428	3.930	4.318
13	0.694	0.870	1.079	1.350	1.771	2.160	2.282	2.650	3.012	3.372	3.852	4.221
14	0.692	0.868	1.076	1.345	1.761	2.145	2.264	2.624	2.977	3.326	3.787	4.140
15	0.691	0.866	1.074	1.341	1.753	2.131	2.249	2.602	2.947	3.286	3.733	4.073
16	0.690	0.865	1.071	1.337	1.746	2.120	2.235	2.583	2.921	3.252	3.686	4.015
17	0.689	0.863	1.069	1.333	1.740	2.110	2.224	2.567	2.898	3.222	3.646	3.965
18	0.688	0.862	1.067	1.330	1.734	2.101	2.214	2.552	2.878	3.197	3.610	3.922
19	0.688	0.861	1.066	1.328	1.729	2.093	2.205	2.539	2.861	3.174	3.579	3.883
20	0.687	0.860	1.064	1.325	1.725	2.086	2.197	2.528	2.845	3.153	3.552	3.850
21	0.686	0.859	1.063	1.323	1.721	2.080	2.189	2.518	2.831	3.135	3.527	3.819
22	0.686	0.858	1.061	1.321	1.717	2.074	2.183	2.508	2.819	3.119	3.505	3.792
23	0.685	0.858	1.060	1.319	1.714	2.069	2.177	2.500	2.807	3.104	3.485	3.768
24	0.685	0.857	1.059	1.318	1.711	2.064	2.172	2.492	2.797	3.091	3.467	3.745
25	0.684	0.856	1.058	1.316	1.708	2.060	2.167	2.485	2.787	3.078	3.450	3.725
26	0.684	0.856	1.058	1.315	1.706	2.056	2.162	2.479	2.779	3.067	3.435	3.707
27	0.684	0.855	1.057	1.314	1.703	2.052	2.158	2.473	2.771	3.057	3.421	3.690
28	0.683	0.855	1.056	1.313	1.701	2.048	2.154	2.467	2.763	3.047	3.408	3.674
29	0.683	0.854	1.055	1.311	1.699	2.045	2.150	2.462	2.756	3.038	3.396	3.659
30	0.683	0.854	1.055	1.310	1.697	2.042	2.147	2.457	2.750	3.030	3.385	3.646
40	0.681	0.851	1.050	1.303	1.684	2.021	2.123	2.423	2.704	2.971	3.307	3.551
50	0.679	0.849	1.047	1.299	1.676	2.009	2.109	2.403	2.678	2.937	3.261	3.496
60	0.679	0.848	1.045	1.296	1.671	2.000	2.099	2.390	2.660	2.915	3.232	3.460
80	0.678	0.846	1.043	1.292	1.664	1.990	2.088	2.374	2.639	2.887	3.195	3.416
100	0.677	0.845	1.042	1.290	1.660	1.984	2.081	2.364	2.626	2.871	3.174	3.390
1000	0.675	0.842	1.037	1.282	1.646	1.962	2.056	2.330	2.581	2.813	3.098	3.300
Z*	0.674	0.842	1.036	1.282	1.645	1.960	2.054	2.326	2.576	2.807	3.090	3.291

Selection of values from R command: qf(0.05, df1, df2, lower.tail = FALSE)

F distribution for p = 0.05

For various pairs of degress of freedom (v_1,v_2), the tabled entries represent the critical values of F above which the proportion p = 0.05 of the distribution falls. (Example, for df = 3,8 and F = 2.92 is exceeded by p = 0.05 of the population.



***p* = 0.05**

	Degrees of freedom for the numerator v_1																	
v_2	1	2	3	4	5	6	7	8	9	10	12	15	20	30	40	60	120	∞
1	161.446	199.499	215.707	224.583	230.160	233.988	236.767	238.884	240.543	241.882	243.905	245.949	248.016	250.096	251.144	252.196	253.254	254.317
2	18.513	19.000	19.164	19.247	19.296	19.329	19.353	19.371	19.385	19.396	19.412	19.429	19.446	19.463	19.471	19.479	19.487	19.496
3	10.128	9.552	9.277	9.117	9.013	8.941	8.887	8.845	8.812	8.785	8.745	8.703	8.660	8.617	8.594	8.572	8.549	8.526
4	7.709	6.944	6.591	6.388	6.256	6.163	6.094	6.041	5.999	5.964	5.912	5.858	5.803	5.746	5.717	5.688	5.658	5.628
5	6.608	5.786	5.409	5.192	5.050	4.950	4.876	4.818	4.772	4.735	4.678	4.619	4.558	4.496	4.464	4.431	4.398	4.365
6	5.987	5.143	4.757	4.534	4.387	4.284	4.207	4.147	4.099	4.060	4.000	3.938	3.874	3.808	3.774	3.740	3.705	3.669
7	5.591	4.737	4.347	4.120	3.972	3.866	3.787	3.726	3.677	3.637	3.575	3.511	3.445	3.376	3.340	3.304	3.267	3.230
8	5.318	4.459	4.066	3.838	3.688	3.581	3.500	3.438	3.388	3.347	3.284	3.218	3.150	3.079	3.043	3.005	2.967	2.928
9	5.117	4.256	3.863	3.633	3.482	3.374	3.293	3.230	3.179	3.137	3.073	3.006	2.936	2.864	2.826	2.787	2.748	2.707
10	4.965	4.103	3.708	3.478	3.326	3.217	3.135	3.072	3.020	2.978	2.913	2.845	2.774	2.700	2.661	2.621	2.580	2.538
11	4.844	3.982	3.587	3.357	3.204	3.095	3.012	2.948	2.896	2.854	2.788	2.719	2.646	2.570	2.531	2.490	2.448	2.404
12	4.747	3.885	3.490	3.259	3.106	2.996	2.913	2.849	2.796	2.753	2.687	2.617	2.544	2.466	2.426	2.384	2.341	2.296
13	4.667	3.806	3.411	3.179	3.025	2.915	2.832	2.767	2.714	2.671	2.604	2.533	2.459	2.380	2.339	2.297	2.252	2.206
14	4.600	3.739	3.344	3.112	2.958	2.848	2.764	2.699	2.646	2.602	2.534	2.463	2.388	2.308	2.266	2.223	2.178	2.131
15	4.543	3.682	3.287	3.056	2.901	2.790	2.707	2.641	2.588	2.544	2.475	2.403	2.328	2.247	2.204	2.160	2.114	2.066
16	4.494	3.634	3.239	3.007	2.852	2.741	2.657	2.591	2.538	2.494	2.425	2.352	2.276	2.194	2.151	2.106	2.059	2.010
17	4.451	3.592	3.197	2.965	2.810	2.699	2.614	2.548	2.494	2.450	2.381	2.308	2.230	2.148	2.104	2.058	2.011	1.960
18	4.414	3.555	3.160	2.928	2.773	2.661	2.577	2.510	2.456	2.412	2.342	2.269	2.191	2.107	2.063	2.017	1.968	1.917
19	4.381	3.522	3.127	2.895	2.740	2.628	2.544	2.477	2.423	2.378	2.308	2.234	2.155	2.071	2.026	1.980	1.930	1.878
20	4.351	3.493	3.098	2.866	2.711	2.599	2.514	2.447	2.393	2.348	2.278	2.203	2.124	2.039	1.994	1.946	1.896	1.843
21	4.325	3.467	3.072	2.840	2.685	2.573	2.488	2.420	2.366	2.321	2.250	2.176	2.096	2.010	1.965	1.916	1.866	1.812
22	4.301	3.443	3.049	2.817	2.661	2.549	2.464	2.397	2.342	2.297	2.226	2.151	2.071	1.984	1.938	1.889	1.838	1.783
23	4.279	3.422	3.028	2.796	2.640	2.528	2.442	2.375	2.320	2.275	2.204	2.128	2.048	1.961	1.914	1.865	1.813	1.757
24	4.260	3.403	3.009	2.776	2.621	2.508	2.423	2.355	2.300	2.255	2.183	2.108	2.027	1.939	1.892	1.842	1.790	1.733
30	4.171	3.316	2.922	2.690	2.534	2.421	2.334	2.266	2.211	2.165	2.092	2.015	1.932	1.841	1.792	1.740	1.683	1.622
40	4.085	3.232	2.839	2.606	2.449	2.336	2.249	2.180	2.124	2.077	2.003	1.924	1.839	1.744	1.693	1.637	1.577	1.509
60	4.001	3.150	2.758	2.525	2.368	2.254	2.167	2.097	2.040	1.993	1.917	1.836	1.748	1.649	1.594	1.534	1.467	1.389
120	3.920	3.072	2.680	2.447	2.290	2.175	2.087	2.016	1.959	1.910	1.834	1.750	1.659	1.554	1.495	1.429	1.352	1.254
∞	3.841	2.996	2.605	2.372	2.214	2.099	2.010	1.938	1.880	1.831	1.752	1.666	1.571	1.459	1.394	1.318	1.221	1.000

TABLE XIV Critical Values of T_L and T_U for the Wilcoxon Rank Sum Test: Independent Samples

Test statistic is the rank sum associated with the smaller sample (if equal sample sizes, either rank sum can be used).

a. $\alpha = .025$ one-tailed; $\alpha = .05$ two-tailed

n_2 \ n_1	3		4		5		6		7		8		9		10	
	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U
3	5	16	6	18	6	21	7	23	7	26	8	28	8	31	9	33
4	6	18	11	25	12	28	12	32	13	35	14	38	15	41	16	44
5	6	21	12	28	18	37	19	41	20	45	21	49	22	53	24	56
6	7	23	12	32	19	41	26	52	28	56	29	61	31	65	32	70
7	7	26	13	35	20	45	28	56	37	68	39	73	41	78	43	83
8	8	28	14	38	21	49	29	61	39	73	49	87	51	93	54	98
9	8	31	15	41	22	53	31	65	41	78	51	93	63	108	66	114
10	9	33	16	44	24	56	32	70	43	83	54	98	66	114	79	131

b. $\alpha = .05$ one-tailed; $\alpha = .10$ two-tailed

n_2 \ n_1	3		4		5		6		7		8		9		10	
	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U
3	6	15	7	17	7	20	8	22	9	24	9	27	10	29	11	31
4	7	17	12	24	13	27	14	30	15	33	16	36	17	39	18	42
5	7	20	13	27	19	36	20	40	22	43	24	46	25	50	26	54
6	8	22	14	30	20	40	28	50	30	54	32	58	33	63	35	67
7	9	24	15	33	22	43	30	54	39	66	41	71	43	76	46	80
8	9	27	16	36	24	46	32	58	41	71	52	84	54	90	57	95
9	10	29	17	39	25	50	33	63	43	76	54	90	66	105	69	111
10	11	31	18	42	26	54	35	67	46	80	57	95	69	111	83	127

Source: From F. Wilcoxon, and R. A. Wilcox. "Some Rapid Approximate Statistical Procedures," 1964, 20–23. Courtesy of Lederle Laboratories Division of American Cyanamid Company, Madison, NJ.